

QA Test Case Suite

Functional Testing for Piano Beats & Cover Fire

1. Piano Beats (Rhythm Game)

ID	Test Scenario	Test Steps	Expected Result
PB-TC-01	Single Tap Accuracy	Tap a single note precisely when it hits the bar.	Score registers as 'Perfect'.
PB-TC-02	Long Note Holding	Press and hold a long note slider until the end.	Continuous score accumulation without breaking combo.
PB-TC-03	Multi-Touch Sync	Tap two simultaneous notes appearing at once.	Both notes register hits correctly.
PB-TC-04	Audio Latency Adjustment	Change latency settings in the options menu.	Note visual arrival aligns with audio timing changes.
PB-TC-05	Combo Multiplier	Achieve 50 consecutive 'Perfect' hits.	Score multiplier increases (e.g., 2x, 4x).
PB-TC-06	Song Interruption	Receive a phone call during gameplay.	Game automatically pauses; audio stops.
PB-TC-07	Low Battery Warning	Play song until device hits 5% battery.	System popup appears; game handles frame drop gracefully.
PB-TC-08	In-App Purchase	Unlock a premium song using virtual currency.	Currency deducted; song becomes playable immediately.
PB-TC-09	Leaderboard Update	Complete a song with a new high score.	Global leaderboard updates within 60 seconds.
PB-TC-10	Calibration Tool	Run the manual sync calibration tool.	System accurately calculates and saves MS offset.

2. Cover Fire (Shooter Game)

ID	Test Scenario	Test Steps	Expected Result
CF-TC-01	Weapon Reloading	Fire all bullets and trigger auto-reload.	Reload animation plays; magazine count resets.
CF-TC-02	Headshot Physics	Aim and fire at the head of an enemy NPC.	One-shot kill registered with specific 'Headshot' icon.
CF-TC-03	Cover Mechanism	Move character behind a concrete barrier.	Character enters 'Cover' state; hit damage reduced/negated.
CF-TC-04	Grenade Splash Damage	Throw grenade into a cluster of 3 enemies.	All enemies in radius take damage based on distance.
CF-TC-05	Weapon Switch	Quick-swap from Sniper to Pistol.	Switch happens within defined frame window; no animation lag.
CF-TC-06	Enemy AI Pathing	Trigger alarm and observe enemy movement.	AI takes cover and flank player logically.
CF-TC-07	Zoom Scope Clarity	Use long-range scope on high-res device.	Texture remains sharp; no pixelation or frame stutter.
CF-TC-08	Mission Failure	Allow player health to reach zero.	'Mission Failed' screen appears with 'Restart' option.
CF-TC-09	Offline Playback	Launch the game with Airplane Mode enabled.	Campaign missions playable; online features grayed out.
CF-TC-10	Environmental Destructibility	Shoot a fuel barrel near enemies.	Barrel explodes causing environmental damage to surroundings.